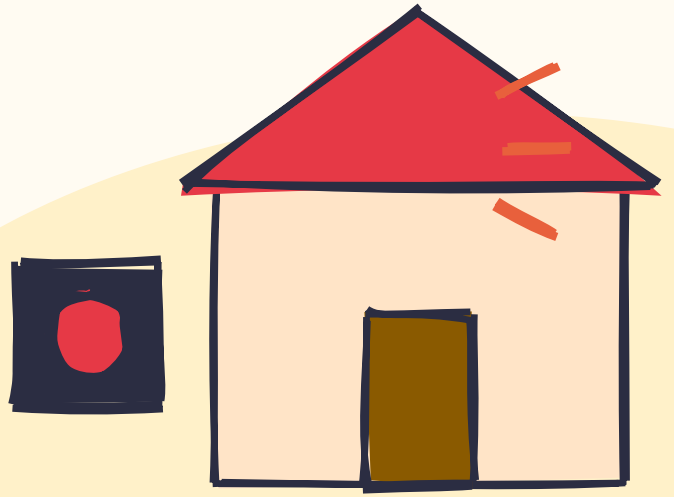
Little Explorers · Class 2

Push-Button Doorbell

Ding dong! Who is at the door?

Button | Buzzer | Flashing light



Did you know?
The first electric doorbells were invented about 200 years ago. Before that, people knocked or pulled a rope!

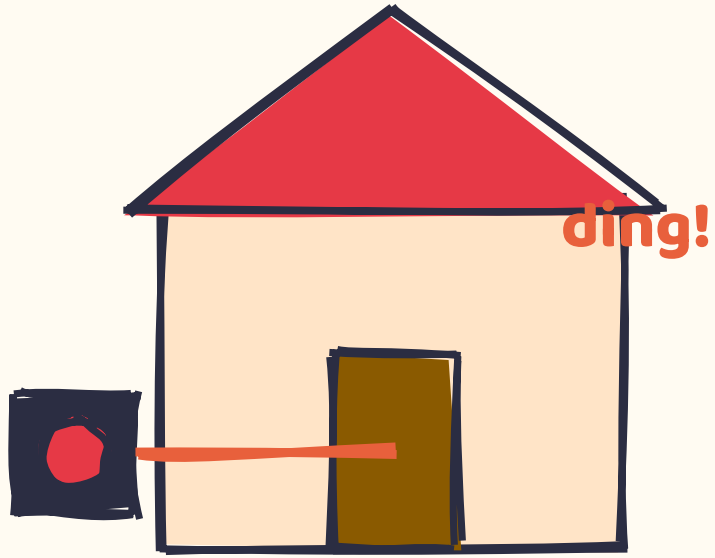
LEARN

Let's wonder...

How does a button make a sound inside?

Press... ding dong!

A friend presses a small button **outside** the door. A bell rings **inside** the house. How does the message get through the wall?



Magic?

A tiny person?

A wire!

Today: we build our own doorbell with a **button**, a **buzzer** and a **light**!

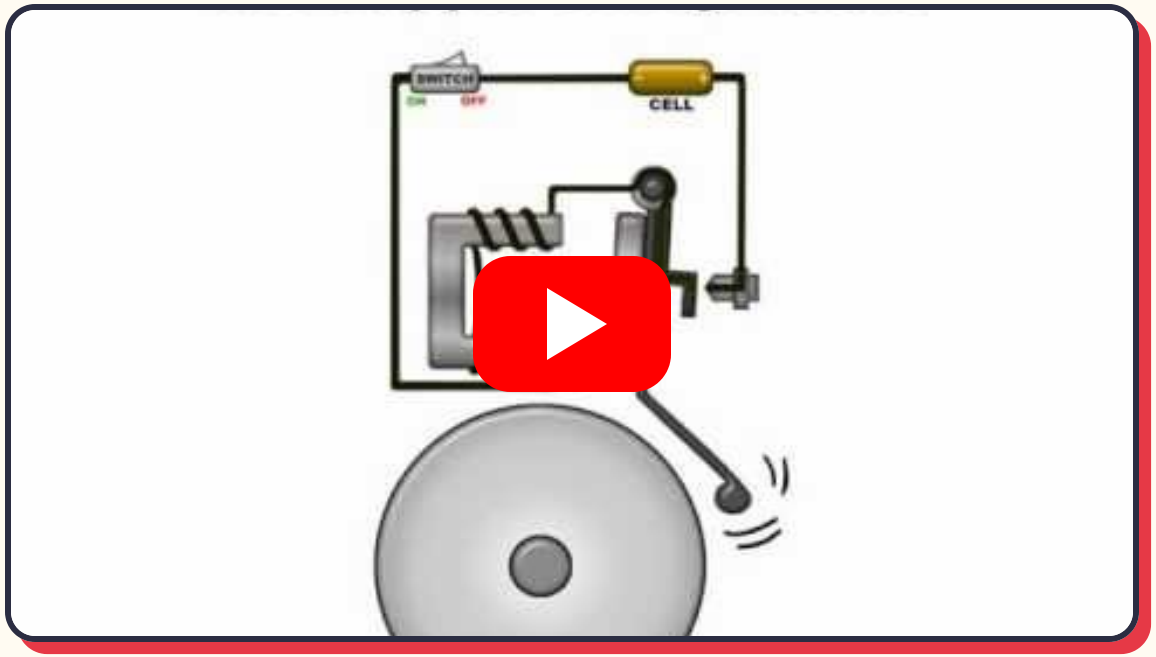
Learning objectives

Today I can…

1	**Press:** explain that pressing a **button** closes a circuit	**Proof:** show a friend the button and the beep
2	**Input, output:** say which part is the **input** and which is the **output**	**Proof:** point to the button (input) and the buzzer (output)
3	**For everyone:** add a **flashing light** so everyone knows a visitor is here	**Proof:** my bell beeps AND flashes

Objective 1 · Press · video

Video time!



How an Electric Bell Works

Bodhaguru · about 5 minutes · youtu.be/HpTp227586s

Watch for...

1. What do you **press** to make the bell ring?
2. What happens when you **let go**?
3. Where have you heard a bell like this?

Scan to watch

After the video: Press an imaginary button on your partner's hand. They say **ding dong!** Swap!

Objective 1 · Press

A button is a little bridge

The idea

Electricity needs a full **circle** (a circuit) to flow.

Not pressed: there is a gap. No electricity, no sound.
Pressed: the button closes the gap. Electricity flows. **Beep!**

Let go
Let go of the button and the bridge goes up again. The sound stops!

Example: a bridge over a river



bridge down = cars cross!

When the bridge is **up**, cars cannot cross. When the bridge comes **down**, cars drive across. Pressing the button puts the bridge down for the electricity!

LEARN

Objective 2 · Input, output

Input in, output out

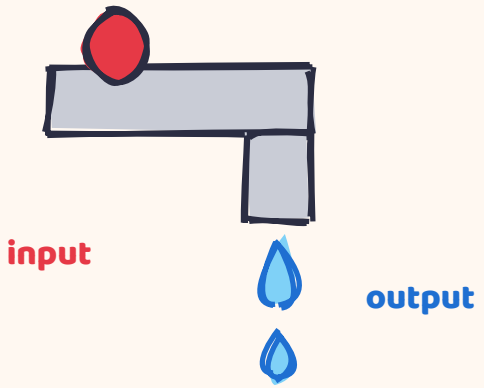
The idea

Machines have **inputs** and **outputs**.

Input = how we tell the machine something (the **button**).
Output = what the machine does back (the **beep** and the **light**).

Think
Can you name the input and output of a TV remote?

Example: a water tap



Turning the tap handle is the **input**. Water coming out is the **output**. Turn it more, more water comes out!

Buttons everywhere

Where do we press buttons?



Lift

Press a number, the lift goes to that floor



Microwave

Press start, it heats your food



Game controller

Press A, your character jumps



Light switch

Press it, the room lights up

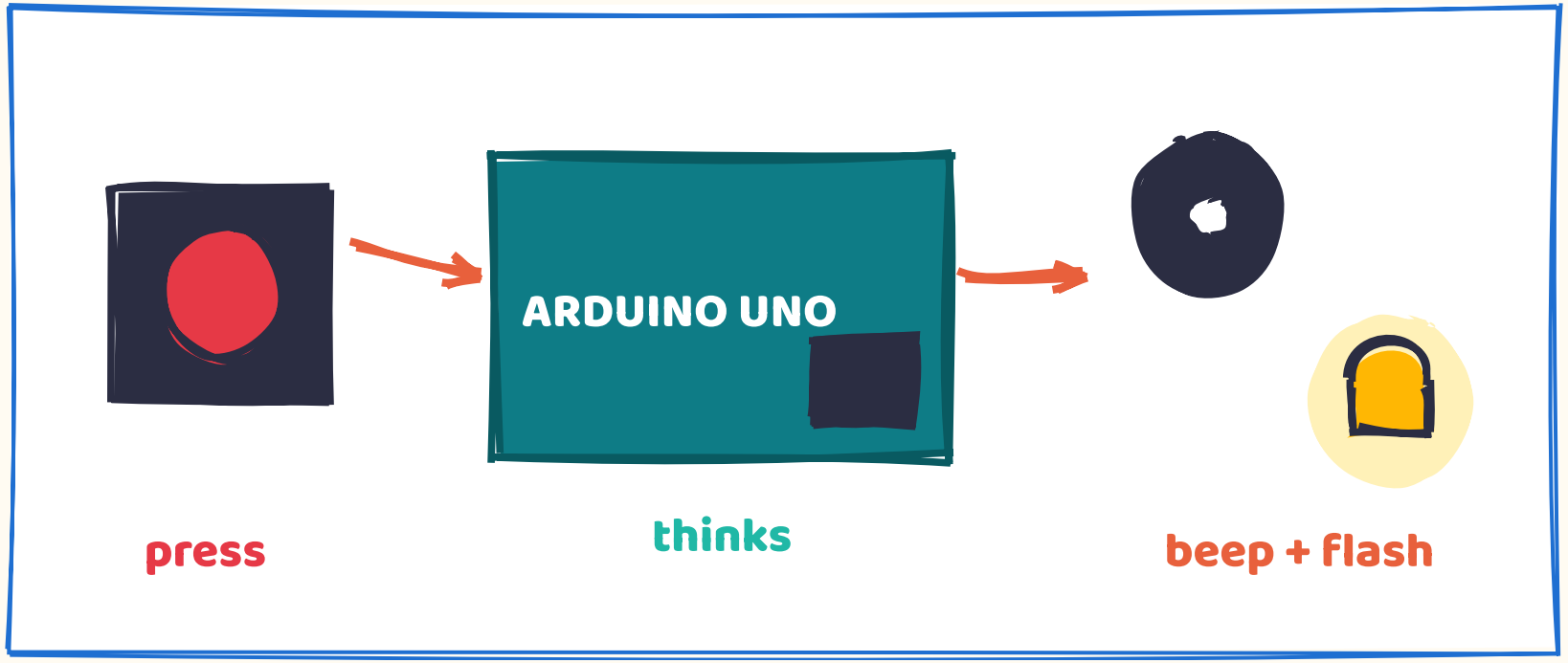
? How many buttons can you find in our classroom?

Count them

Look for buttons at home tonight. Who found the most?

Our machine

Feel, think, show



How it works

The **button** tells the Arduino someone pressed.
The Arduino **thinks**: was it pressed?
The **buzzer** beeps and the **light** flashes.

Teacher: parts are pre-wired on the breadboard. Children plug in the USB cable, snap the picture blocks in mBlock and test.

Objective 3 · For everyone

A doorbell for everyone

The idea

Some people **cannot hear** sounds very well. A beeping bell does not help them.

We add a **flashing light**. Now people who cannot hear will **see** that someone is at the door!

Kind engineers
Engineers think about ALL the people who will use their invention.

Example: fire alarms



see it AND hear it

In schools and hotels, fire alarms **beep loudly** and **flash bright lights**. Everyone gets the message, whether they can hear or not.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human doorbell
2. Snap the picture blocks
3. How long should it ring?
4. Doorbell challenges

MAKE

Class game

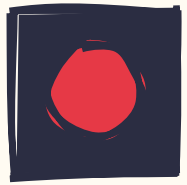
Human doorbell

How to play

1. One child is the **button**: hold out a hand.
2. One child is the **bell**.
3. When someone touches the button's hand, the bell sings **ding dong!**
4. Let go = silence.
5. Now add a **light** child who flashes their hands too!

Then discuss

Which child is the **input**? Which children are the **outputs**?



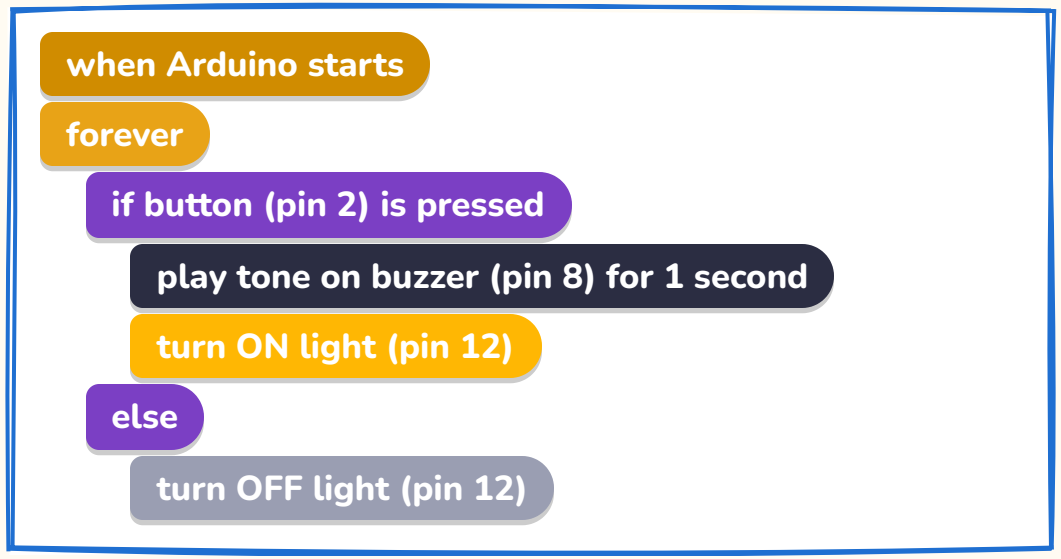
Big idea

Every machine we build has an input and an output, just like our human doorbell.

MAKE

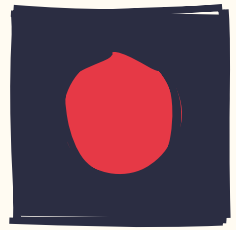
Time to build!

Snap the picture blocks



Read it like a story

IF the button is pressed: beep and light up.
ELSE: light off.
Then check again, again, again!



Forever?
The Arduino checks the button again and again, very fast, all day long.

MAKE

Experiment!

How long should it ring?

Change the time to…	It felt…	We liked it?
1 second	short / long?	😀 / 😐
2 seconds	short / long?	😀 / 😐
5 seconds	short / long?	😀 / 😐

Talk about it

Which ring time is **best** for a doorbell? Too short and nobody hears it. Too long and it is annoying!

Change one number

Change only the number in the blue block. Press the button and listen.

MAKE

Level up!

Doorbell challenges

Choose your challenge



Ding dong: play two notes, high then low, like a real bell.



Party light: make the light blink 3 times.



Secret knock: make a different sound if the button is pressed twice.



Quiet time: a soft beep for a sleeping baby.

Think like an engineer

Where else could a button and a light help someone?

Boss level

Make the bell play a tune you know!

Objective check

Can I really do it? Prove it.

01	Show your doorbell. What happens when you press the button?	○ ○ ○
02	Point to the **input**. Point to the **output**.	○ ○ ○
03	Who is the flashing light for?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Ask children to explain to a partner, then to the class. Picture cards help: button, buzzer, light.

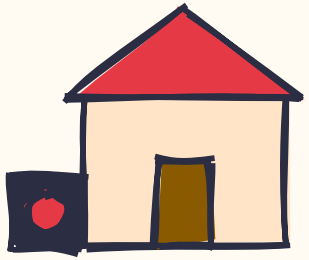
Show what you know

What did we learn?

- [x] A button closes the circuit like a bridge
- [x] The button is the input
- [x] The beep and light are outputs
- [x] A light helps people who cannot hear

At home

Find 3 buttons at home. What does each one do?



Exit ticket: what is the input of our doorbell?

Next class: make a Lemon Battery lamp!